HDOC Day-58

Question: Write a C program to read a variable-length integer array and k. Pass the array by address to kth_smallest(), sort it in ascending order using bubble sort, and print the kth smallest element. Count duplicates separately and print -1 for invalid k.

C Program

```
#include <stdio.h>

void kth_smallest(int *arr, int size, int k) {
    int i, j, temp;

    if (k <= 0 || k > size) {
        printf("-1\n");
        return;
    }

    for (i = 0; i < size - 1; i++) {
        for (j = 0; j < size - i - 1; j++) {
            if (*(arr + j) > *(arr + j + 1)) {
                temp = *(arr + j);
                *(arr + j) = *(arr + j + 1);
                *(arr + j + 1) = temp;
            }
        }
    }

    printf("%d\n", *(arr + k - 1));
}

int main(void) {
    int size, k, i;
    scanf("%d", &size);
    if (size <= 0) {
        printf("-1\n");
        return 0;
    }

    int arr[size];
    for (i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
    scanf("%d", &k);

    kth_smallest(arr, size, k);
    return 0;
}
```

Sample Input

```
6
7 2 4 2 9 1
3
```

Sample Output

```
2
```

Explanation: After bubble sort, the array is 1, 2, 2, 4, 7, 9. The third element is 2. Repeated values are counted separately. For k = 0 or k greater than the size, the output is -1.